JUNSU KIM

Curriculum Vitae

sam9732@kaist.ac.kr | Homepage | LinkedIn | GitHub

Research Interests

- Egocentric and embodied AI for tool use, physical interactions, and action-driven state changes.
- 3D/4D scene understanding and action-centric world models trained on human and robot data.

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
B.S. in Electrical Engineering — Mar 2021 – *Expected Aug 2027*
GPA: 4.09/4.3 (Major GPA: 4.19/4.3)
Military Service, Republic of Korea Army: Jul 2022 – Jan 2024

Nanyang Technological University (NTU) — Singapore
Exchange Student — Jan 2026 – May 2026
School of Electrical and Electronic Engineering

Publications

[1] **EgoTools: Towards Tool-Centric Reasoning in Real-World Egocentric Videos**.
[Project Page] [Paper].
Shulin Tian*, **Junsu Kim***, Shuai Liu*, Hao Li*, ..., Ziwei Liu.
EMNLP 2026 *(Main Conference)*.

[2] **C3G: Learning Compact 3D Representations with 2K Gaussians**.
[Project Page] [Paper].
Honggyu An*, Jaewoo Jung*, ..., **Junsu Kim**, ..., Seungryong Kim.
CVPR 2026.

[3] **DreamX: Video Diffusion Model Is A Universal Latent Action Learner**.
Jisoo Kim*, Chaehyun Kim*, **Junsu Kim**, ..., Seungryong Kim.
Under review at ICLR 2027.

*Equal contribution.

Research Experience

Maum AI, WoRV Division, Future Intelligence Lab — Aug 2026 – Present
Research Intern

- Lead task design for a long-horizon **human–robot collaboration benchmark**, developing simulated cooking scenarios, physical success checks, and partial-information variants; built an early state-engine prototype and found that symbolic task constraints masked visual activity-recognition errors.

S-Lab (MMLab), NTU — Jan 2026 – Present
Research Intern (Advisor: Prof. Ziwei Liu, NTU)

- Co-led **EgoTools**, a dataset and benchmark for tool-centric reasoning: collected ∼100 hours of egocentric tool-use video and designed a 1K-QA benchmark across four reasoning tracks, with annotation and quality-control workflows.

- Ran input ablations validating the benchmark's reliance on visual and temporal evidence; contributed to the Qwen3-VL-8B fine-tuning study (EgoTools-Bench **50.0%** → **60.9%**) and its cross-benchmark generalization analysis.

Computer Vision Lab (CVLab), KAIST AI Jun 2025 – Present
Research Intern (Advisor: Prof. Seungryong Kim, KAIST AI)

- *Jun 2026 – Oct 2026:* For **DreamX**, ran the latent-action baseline comparison on LARYBench under a shared protocol, and co-ran human-to-robot action-transfer experiments on a bimanual robot, recording human demonstrations and paired teleoperated reference trajectories.
- *Jun 2025 – Nov 2025:* Contributed to **C3G**, analyzing DINOv2-based attention maps to study how compact Gaussian queries aggregate multi-view information.

Video and Image Computing Lab (VICLab), KAIST Dec 2024 – Jun 2025
Research Intern (Advisor: Prof. Mun Churl Kim, KAIST EE)

- Implemented and evaluated object-detection pipelines spanning two-stage, YOLO-style, and Vision Transformer architectures for few-shot and anomaly detection tasks.

Honors & Scholarships

KAIST **Presidential Fellowship** (15th) Feb 2025 – Present
National Science & Engineering Scholarship for Academic Excellence Aug 2025 – Present
Woon Hae Scholarship (12th) Feb 2025 – Dec 2025
KAIST **Dean's List**, College of Engineering Fall 2021, Spring 2024
awarded to the top 3% of students
KAIST Shared Lives Award (Top-ranked Recipient) Feb 2026
For exceptional and sustained blood donation service

Leadership & Service

President, EERun (KAIST EE Running Club) Sep 2024 – Dec 2025
Organized weekly training sessions and semester events for KAIST EE students.
Google Student Ambassador Sep 2025 – Dec 2025
Promoted Gemini Pro to KAIST students through short-form video content.
Additional Activities 2022 – Present
KAIST AI Studying Club, Young Engineers Honor Society (YEHS), SilverLining, Freshman Proctor.

Skills

Programming C/C++, Python, PyTorch, Git, LaTeX
Research Tools video preprocessing, dataset curation, annotation pipelines, benchmark construction, evaluation scripts
Languages Korean (native), English (fluent, iBT TOEFL 110)